



MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

BSE4104: EMERGING TRENDS IN SOFTWARE ENGINEERING

AI-NATIVE & AGENTIC ENGINEERING CAPSTONE: WEEK 3 SUBMISSION

WEEK 3 RAG ARCHITECTURE

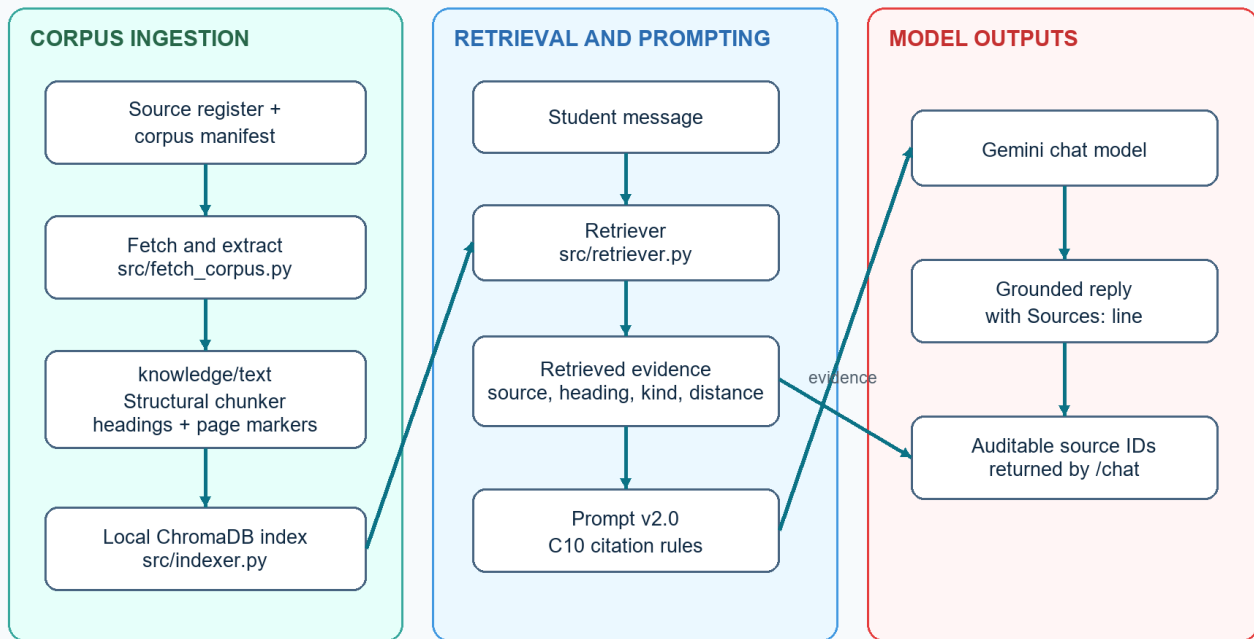
PROJECT: UNIVERSITY STUDENT-SUPPORT CASE AGENT

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Week 3 RAG Architecture

Week 3 Retrieval-Augmented Generation Architecture



Solid arrows show data flow. The index is local; Gemini is used when the configured provider is selected.

Data flow

- 1 src/fetch_corpus.py fetches public sources, records checksums and provenance in knowledge/corpus.json, and writes extracted text under knowledge/text/.
- 2 src/chunker.py splits text on headings and PDF page markers, packs paragraphs into bounded overlapping chunks, and preserves document metadata.
- 3 src/indexer.py embeds and persists chunks in knowledge/index/ using ChromaDB.
- 4 src/retriever.py queries the index and formats a RETRIEVED EVIDENCE block. src/main.py also returns the retrieved document IDs in the API response.
- 5 prompts/v2.0.md instructs the chat model to answer only from evidence, acknowledge unsupported questions, and include source citations.

The index is local, but the configured Gemini embedding and chat providers send query or model text to Google when selected. Synthetic documents D08 and D09 are labelled in the corpus metadata and source register.

Current limitation

The retrieval path now applies an evidence-coverage gate and unsupported-request filter before evidence is passed to the model. Short, specific documents receive metadata-aware reranking. Several real sources are image-only scans and remain excluded until OCR is run. See the evaluation and failure catalogue for the measured impact.